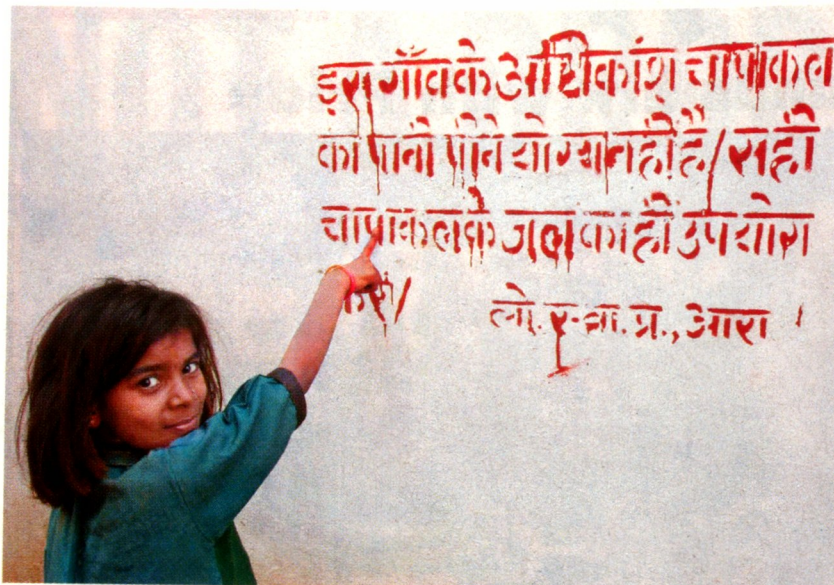




Pradesh, Nepal and Assam, you would actually be touching the flood plains of Ganga-Meghna-Brahmaputra belt," says Ashok Kumar Ghosh, professor at the department of environment and water management, at AN College, Patna.

Scientific evidence suggests that naturally-occurring arsenic has travelled down from the Himalayas and seeped into the soil of the plains. Researchers believe that Himalayan rivers carry sediments that are rich in arsenic, which makes it easy for the metalloid to seep into the groundwater along the rivers' course.

Arsenic was there right from the beginning, but it was present in a composite form, which was insoluble in water. "With excessive consumption of groundwater and some chemical changes, the insoluble arsenic has got converted into a soluble component. This has increased its level in water," says Ghosh.

The study further establishes that older tubewells have greater chances of contamination. A similar pattern of tubewell contamination has also been observed in West Bengal and Bangladesh. "There is a health pandemic, but our system is fractured," says an NGO official. The village-level health officials are not concerned about the quality of water provided to the villagers. "We need to establish a link between arsenic cleaning and the cleaning of our water

TRAGIC: A mother points out the ill-effects of arsenic in Ara

THE KILLER DEPOSIT

- Researchers believe rivers originating from the Himalayas carry sediments that are rich in arsenic. The metalloid seeps into the groundwater along the rivers' trajectory.
- Other reasons behind arsenic poisoning include oxidation and excessive withdrawal of groundwater.
- The best way to mitigate the poisoning is by creating awareness among people and using alternative sources of water.



POISONED CHALICE: A warning about contaminated water in Ara, Bhojpur sources. Only then can we avail fruitful results," adds the official.

Taking cue from the seriousness of the issue, the state PHED has stepped forward and established several water testing laboratories, not only in Patna but also in other arsenic-affected districts of the state. Besides, the department has outsourced water testing jobs to private experts. Currently, 2.25 lakh water sources in the state are being examined. Once the department finds concrete results, it will draw up plans to ensure that freshwater supplies are made available to villagers by cleaning the affected tubewells properly, so that no one is forced to drink contaminated water. The department is also ensuring that water testing kits are available at all panchayats in the state. The process is likely to conclude by April next year.

With the help of the United Nations Children's Fund, (UNICEF), PHED has mapped water sources and marked them with different colours to indicate the potability of water flowing through them. For instance, arsenic-free tubewells have been painted blue, while the ones that are polluted have been marked red. In addition, people are being informed about unsafe water sources so that they can keep the problems at bay. With these models proving to be successful, the government has allocated Rs 107 crore to expand these efforts to all affected villages and schools with technical assistance from UNICEF.

Arsenic contamination first emerged as a major health issue in the 1980s, with the documentation of poisoned communities in Bangladesh and West Bengal. A 2003 survey in Semria Ojha Patti, a village in Bihar, had brought into notice the escalating problem of groundwater poisoning and the health problems arising due to it.

The latest PHED survey, however, has established that the problem could be more widespread than presumed and might affect the middle and upper Gangetic plains as well. This should surely ring alarm bells in other states.